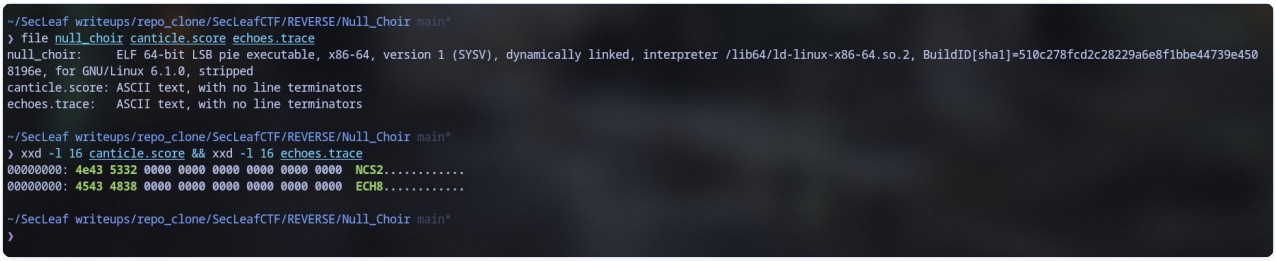
REVERSE HARD (500 PTS)

Null Choir

Binary Triage & Components

Authored by 5h1kh4r. Challenge bundle:

- `null_choir` : Stripped ELF64 binary.
- `canticle.score` : 196,864-byte binary blob (magic `NCS2`).
- `echoes.trace` : 3,104-byte binary blob (magic `ECH8`).



Binary Triage

The binary is a custom register-based virtual machine interpreter. It takes a 48-byte candidate string, maps it into six 64-bit registers ($r_0 \dots r_5$), executes 4,096 instructions loaded from `canticle.score`, and compares the end state against an encrypted target buffer in `.rodata`.

- Output on success: `Harmony.` (exit code 0).
- Output on failure: `Dissonance.` (exit code 1).

Trace Authentication & Conductor State

`echoes.trace` contains 192 candidate records (16 bytes each) meant to seed the VM. Each record layout: `'E' 'C'` + 8-byte value + 4-byte FNV-1a checksum.
Filtering rules extracted from binary: 1. Header matches `EC\xe5` (`tag = (seed >> 40) & 0xff`). 2. Slot index ≤ 31. 3. FNV-1a checksum of the first 12 bytes matches bytes 12..15.

Out of 192 records, exactly 32 are authentic, forming a complete permutation of slots $0..31$. Folding the genuine slot values with SplitMix64 generates the conductor state: `state = 0x866a952e86bb7d8b` Bank selection `(state >> 11) % 3 = 0` points execution to instruction Bank 0 at offset `0x100`.

VM Instruction Set Architecture

- **Registers:** Six 64-bit general-purpose registers $r_0 \dots r_5$.
- **Opening Transform:**

$r_k = \text{rol}(r_k \oplus \text{mix}(\text{state} \oplus (k+1)G), 14 + 7k)$

- **Opcodes (24 distinct operations):**

Linear arithmetic (`ADD` , `SUB` , `XOR`), circular shifts (`ROL` , `ROR`), and SplitMix64 non-linear mixing.

- **Diffusion Layers:**

Every 64 instructions, an inter-register diffusion step executes: $r_k = r_k \oplus r_{(k+1)\%6} \oplus \text{mix}(\dots)$

Reversing Execution

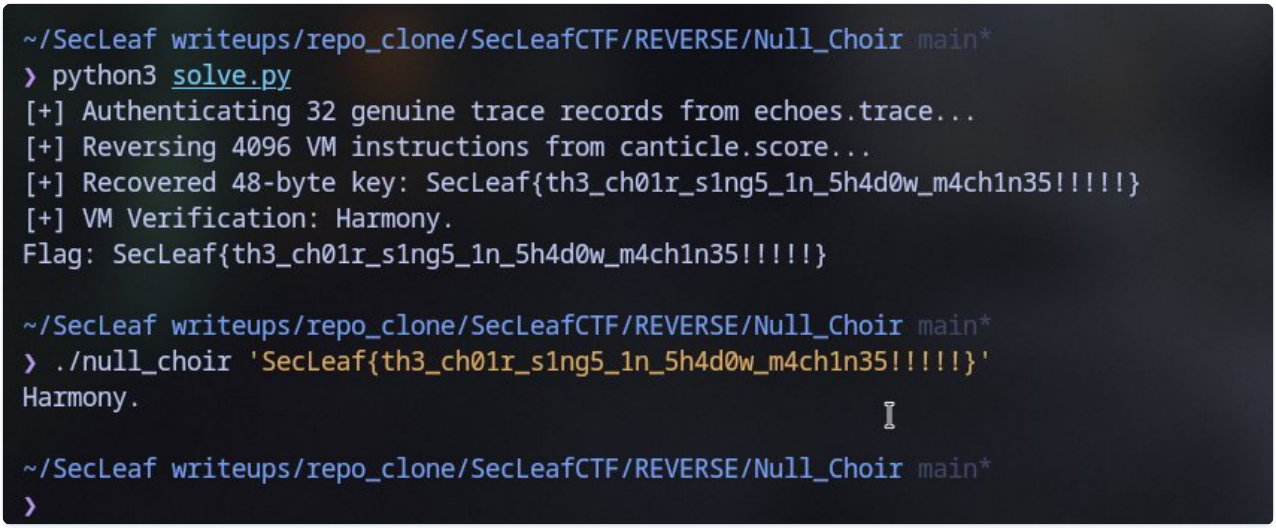
Crucially, every single instruction and diffusion round in the VM is strictly invertible: 1. Extract the 48-byte expected final state from `canticle.score` by XORing with the key stream. 2. Step backwards from instruction 4095 down to 0, applying inverse operations. 3. Invert the 64-step diffusion layers when `i % 64 == 63` . 4. Invert the opening transform to recover the initial 48 bytes.

Solution Script: `Null_Choir.py` (included in attached solution archive)

Verification

```
$ python3 solve.py [+] Authenticating 32 genuine trace records from echoes.trace... [+] Reversing 4096 VM instructions from canticle.score... [+] Recovered 48-byte key: SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!} [+] VM Verification: Harmony. Flag: SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}

$ ./null_choir 'SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}' Harmony.
```



Flag and Interpreter Verification

Flag:

Solution Script: Null_Choir.py (included in attached solution archive)

RECOVERED FLAG **SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}**